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Determinants of Groundwater use in Agriculture: A Micro Level Analysis in Balasore district, Odisha, India

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Groundwater is becoming essential for irrigation due to the lack of surface irrigation infrastructure and limited knowledge about soil health, which has led to its unsustainable extraction. The study aims to find the determinants and their impacts on groundwater extraction. The determinants of groundwater extraction namely, soil testing, seed replacement rate, type of fertilizer used, crop diversification, land size, per-capita food consumption, and financing source, are found to have more impact on groundwater extraction in Balasore district. The OLS regression was used to assess the functional relationship of these variables with the dependent variable. As per the results, soil testing, crop diversification, land size, per-capita food consumption, and sources of finance have significant impacts on groundwater use. To achieve sustainability, awareness of soil nutrients and investment in water conservation infrastructure is suggested.

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